

Juan Carlos Barrera

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SUMMARY

Software engineer with 4 years of production web-fullstack experience, now focused on systems and distributed-systems engineering. Currently completing a Master of Computer Science majoring in Networks & Distributed Systems and Algorithms & Theory; hands-on C work spans multi-process IPC servers, **epoll**-driven event loops, configurable thread-pool scheduling, and POSIX synchronisation.

EDUCATION

University of Sydney

Master of Computer Science (Advanced Entry)

Networks & Distributed Systems / Algorithms & Theory

- Relevant coursework: Data Structures & Algorithms, Systems Programming (C)
- Colfuturo Scholarship recipient, Colombian Government

Sydney, Australia

2026 – Present

Sergio Arboleda University

Bachelor of Finance and International Trade

- Top 1% nationally – Saber Pro 2020 (230/300)
- Colciencias Research Scholarship
- 70% merit scholarship
- GPA 4.40/5.00

Bogotá, Colombia

2017 – 2021

PROJECTS

Concurrent Animation Server | C, POSIX

2026

- Parallelises RPC execution from a worker-thread pool via round-robin; an enqueueer process enforces per-client order through queues, and concurrent execution on a three-process pipeline (connection handler → enqueueer → processor) driven by **epoll** on IPC fds
- Wired five IPC mechanisms — named pipes, anonymous pipes, SIGUSR1/2 handshake, **timerfd** idle timeouts, and **epoll** event queues — with non-blocking I/O
- Built cross-client barrier synchronisation on shared resources; resolved disconnecting deadlocks; AddressSanitizer clean throughout

Sprite Animation Library | C

2026

- Implemented a sprite animation library in C; custom doubly-linked list with O(1) node operations via cached pointers, BMP file parser, and alpha-composited frame renderer — zero external dependencies
- Designed opaque-struct API enforcing strict memory ownership; all allocation and teardown handled by the library, verified leak-free under AddressSanitizer

EXPERIENCE

Fullstack Developer

Jul. 2024 – Jun. 2025 · Dec. 2025 – Present

AdviceFlow

Lisbon, Portugal (Remote)

- Architected backend services coordinating asynchronous data pipelines across multiple third-party APIs under strict latency requirements
- Reduced API response latency through server-side caching strategies and query optimisation on GCP-hosted infrastructure

Fullstack Developer

Nov. 2024 – Feb. 2026

GeneralDev

Denver, United States (Remote)

- Built and shipped production web services with focus on server-side performance, data-fetching efficiency, and caching architecture
- Built headless storefront architecture decoupling rendering from commerce APIs; managed data orchestration, caching, and SSR performance across the full request lifecycle
- Maintained CI/CD pipelines via GitHub Actions; wrote deployment automation and environment configuration scripts

Back End Developer

Mar. 2024 – Jul. 2024

Blackpeak

Vancouver, Canada (Remote)

Back End Senior Developer

Mar. 2022 – Jan. 2024

Castia (YC-backed)

Bogotá, Colombia

TECHNICAL SKILLS

Languages: C, TypeScript, JavaScript, Python, SQL, Bash

Systems & Environment: Unix/Linux (Arch, openSUSE), POSIX, Neovim, Git, Docker, GitHub Actions

Web & Frameworks: Node.js, Next.js, Remix.js, React, FastAPI, Flask

Databases: PostgreSQL, CloudSQL, Supabase

Cloud: Google Cloud Platform (GCP), Vercel